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Arm Solutions at Lightspeed

Enabling Push call back in Fair class

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Why Push task mechanism for EAS ?

- Need to evaluate task placement more often
  - not only at wake up
- Periodic load balance is too big
  - Doesn’t scale w/ tasks
  - Look at CPU granularity
- Periodic load balance is too slow
  - dozens of ms when busy

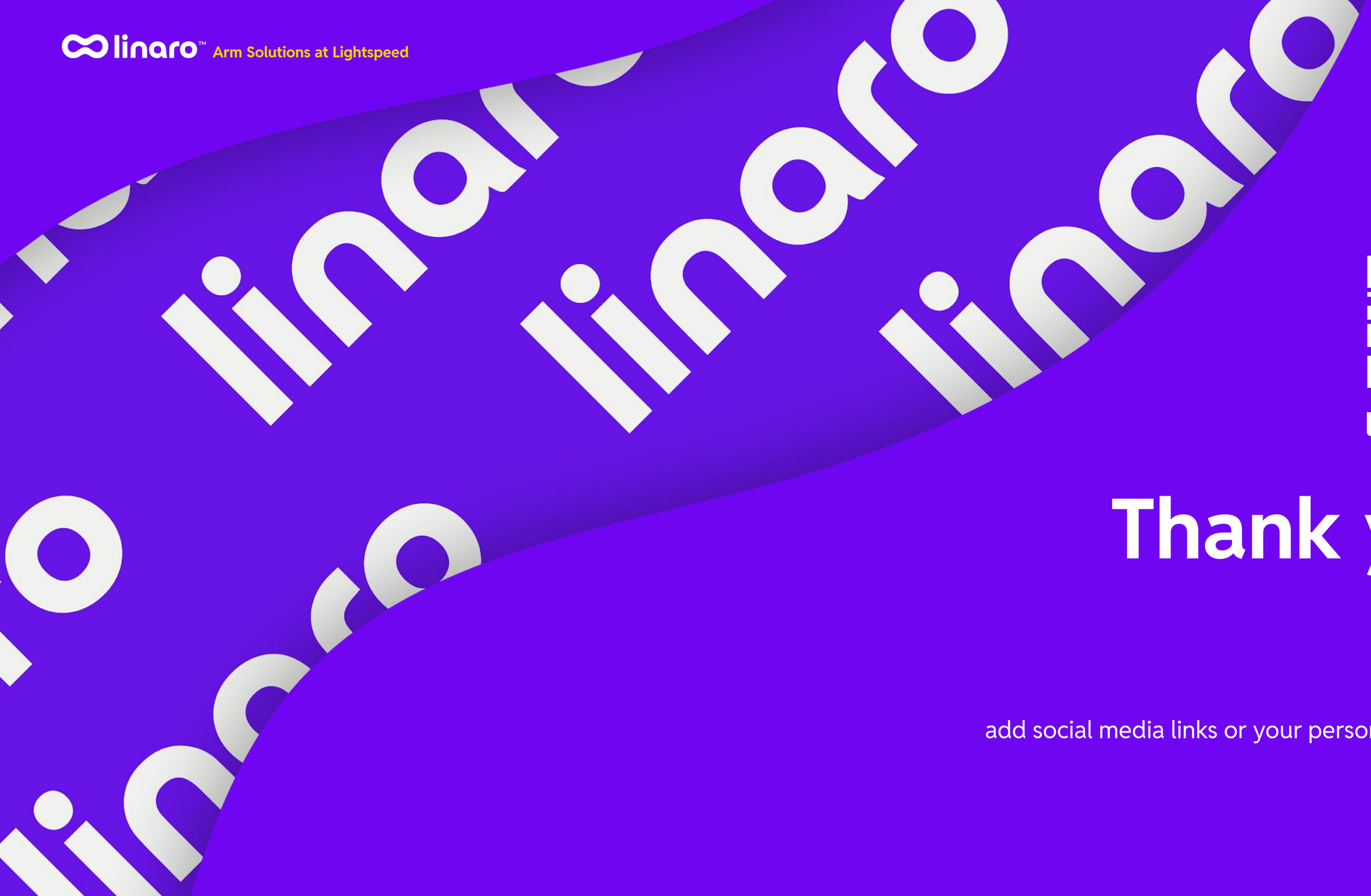
Why same policy everywhere ?

- Use the same selection function everywhere
  - Consolidated decision
  - Can be used outside EAS

- Extend select_task_rq_fair
  - Can be used at anytime
  - can be used when overutilized

Extend Push mechanism for fair

- Periodic load balance is too big
  - can’t be always called
  - skip new idle balance
  - busy period in hundreds of ms
- Per task instead of per CPU
  - take into account new hints like slice



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